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CASE STUDY

H2 TRANSPORT

INDUSTRY USE CASES

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ABSTRACT

Hydrogen is rapidly emerging as a pivotal energy carrier in the global transition towards a zero-emission future. Its versatility allows for diverse applications across transportation and industrial sectors, addressing decarbonization challenges where direct electrification faces limitations. This document highlights real-world examples of hydrogen utilization, from fuel cell electric vehicles (FCEVs) in passenger cars, buses, and heavy-duty trucks, to innovative hydrogen combustion engines in various platforms, and specialized applications in rail, maritime, and industrial logistics.

These case studies demonstrate hydrogen's growing maturity and its crucial role in complementing battery technology to achieve comprehensive emissions reductions.

1. INTRODUCTION

The imperative to decarbonize the global economy has spurred significant innovation in clean energy technologies. Among these, hydrogen stands out for its potential to replace fossil fuels in sectors where emissions are traditionally hard to abate. Whether utilized in fuel cells to generate electricity or directly combusted in modified engines, hydrogen offers a pathway to zero or near-zero emissions at the point of use.

This document presents a survey of tangible, real-world applications of hydrogen technology, illustrating its expanding footprint in creating a sustainable energy landscape.

2. HYDROGEN FUEL CELL ELECTRIC VEHICLES (FCEVS): ROAD-BASED MOBILITY

Hydrogen fuel cells convert hydrogen gas into electricity to power electric motors, with water vapor as the only byproduct. This technology is gaining traction in various road vehicle segments.

2.1 Passenger Cars

Major automotive manufacturers have introduced FCEVs, offering consumers zero-emission driving with ranges and refueling times comparable to conventional vehicles.

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- **Toyota Mirai**: Launched in 2014 as the world's first mass-produced FCEV, the Mirai sedan boasts a range of **400–500 km**, with **over 10,000 units sold globally** [1].
- **Hyundai Nexo**: This hydrogen-powered SUV, launched in 2018, offers similar performance and is available in key markets like California, South Korea, and parts of Europe [1].

While global FCEV sales remain niche (approximately **14,451 units in 2023**, down from **20,700 in 2022** [2]), growth is steady in regions with developing hydrogen refueling infrastructure.

2.2 Buses

Fuel cell buses are increasingly deployed in urban environments to decarbonize public transit and improve air quality.

- **Europe**: By early 2023, **370 hydrogen buses** were operational, with plans for **over 1,200 by 2025** [3]. Notable deployments include London's fleet of 20 hydrogen double-decker buses and significant rollouts in Germany, France, and the Netherlands, often supported by EU initiatives like JIVE, which aims to deploy 300 buses across 22 cities [3].
- **Asia**: China, Japan, and South Korea are leaders in fuel cell bus adoption, with cities like Seoul actively integrating them into their public transport networks. These buses typically offer ranges of **300–400 km** and rapid refueling (around 10 minutes).

2.3 Heavy-Duty Trucks

Hydrogen fuel cells are particularly promising for heavy-duty trucking, where battery solutions often struggle with range, payload capacity, and charging times.

- **Hyundai XCIENT**: Hyundai has deployed **47 XCIENT fuel cell trucks** in Switzerland, accumulating **over 4 million kilometers by 2022**. These trucks offer a ~400 km range with ~35 kg of hydrogen. Hyundai aims for 1,600 units in Switzerland by 2025 and is expanding into other European countries and conducting trials in California with a 30-truck deployment [4].

- **Toyota & Kenworth Partnership**: Successful trials of fuel cell semi-trucks at the Port of Los Angeles, demonstrating ranges over 300 miles, have spurred further development of hydrogen powertrains for freight.
- **H2Accelerate Consortium**: A collaboration between European truck manufacturers (Daimler Truck, Volvo, IVECO) and energy companies (Shell, TotalEnergies, OMV) aims to deploy 150 fuel-cell trucks and associated refueling infrastructure in Europe by the late 2020s. Daimler Truck plans to deliver **5,000 hydrogen trucks by 2030**, supported by Shell's refueling network development [5].

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3. HYDROGEN COMBUSTION ENGINES: A NEW PATHWAY FOR EXISTING TECHNOLOGY

Beyond fuel cells, direct combustion of hydrogen in modified internal combustion engines (ICEs) offers another route to decarbonization, leveraging existing manufacturing expertise.

3.1 Automotive Applications

- **Toyota's hydrogen engine development**: Toyota has actively tested a hydrogen-fueled 1.6L turbocharged engine in a Corolla race car, achieving significant performance improvements and rapid refueling (1.5 minutes). This research has led to a prototype Corolla Cross H2 road car. Toyota has also demonstrated a liquid hydrogen-fueled race car, highlighting the fuel's versatility [6].
- **Cummins hydrogen ICE for trucks**: U.S. engine manufacturer Cummins is developing large-displacement hydrogen engines for heavy vehicles, including a 15-liter engine targeting production in 2027, designed to offer power and torque comparable to diesel counterparts but with **zero CO_2 emissions**. Cummins is collaborating with Daimler (Freightliner) to integrate these engines into popular truck models [7].

3.2 Maritime Applications

Hydrogen and hydrogen-derived fuels like ammonia are being explored for marine engines.

- **Hydroville Ferry (Belgium)**: Launched in 2017, this passenger ferry utilized a modified diesel engine to burn hydrogen (with diesel pilot fuel), serving as an early testbed for marine hydrogen combustion.
- **MAN Energy Solutions**: This major marine engine manufacturer is developing ammonia-fueled engines, with the first delivery planned for 2024 and commercial rollout anticipated by 2027. Ammonia (NH_3) allows for easier storage of hydrogen energy for long-haul shipping.

3.3 Aviation Applications

Modified jet engines can potentially burn hydrogen, eliminating CO_2 emissions.

- **Rolls-Royce & easyJet**: In 2022, Rolls-Royce successfully tested a modern aircraft turbine (AE2100) running on **100% green hydrogen**. They are now advancing to test a full-size Pearl 15 business jet engine with hydrogen at a NASA facility in 2024 [8].
- **Airbus ZEROe**: Airbus is developing concept hydrogen-powered commercial aircraft (ZEROe), targeting entry into service by 2035. These designs envision hydrogen use in either fuel cells for electric propulsion or direct combustion in modified turbine engines.

Hydrogen combustion offers a potentially faster transition for industries reliant on ICEs, though challenges include managing NOx emissions (which can be addressed with aftertreatment) and on-board hydrogen storage.

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4. HYDROGEN IN SPECIALIZED INDUSTRIAL TRANSPORT

Hydrogen is proving effective in various industrial transport modes that are difficult to electrify solely with batteries.

4.1 Hydrogen Trains

Fuel cell trains offer a zero-emission solution for non-electrified or difficult-to-electrify rail lines.

- **Alstom Coradia iLint (Germany)**: Since 2018, Germany has operated these hydrogen passenger trains. In 2022, a rail line in Lower Saxony became **the world's first to be fully powered by 14 hydrogen trains**, replacing diesel units and saving an estimated **4,000 tons of CO_2 annually** [9]. These trains have also been demonstrated in Austria, the Netherlands, and France. The UK (HydroFLEX prototype) and China (hydrogen trams) are also advancing hydrogen rail technology.

4.2 Hydrogen Forklifts and Warehouse Vehicles

Hydrogen fuel cell forklifts represent a significant, albeit less publicized, success story.

- **Large-scale adoption**: Companies like Amazon and Walmart have deployed tens of thousands of hydrogen fuel cell forklifts. Amazon utilized **over 15,000 units by 2022**, with plans for 5,000 more by 2025. Walmart operated **over 9,500 units by 2023** [10]. The key advantages are continuous power delivery throughout a shift and rapid refueling (around 3 minutes), significantly boosting productivity in 24/7 warehouse operations compared to battery-powered alternatives.

4.3 Maritime Transport — Ferries and Ships

The maritime sector is actively experimenting with hydrogen for cleaner operations.

- **MF Hydra (Norway)**: Launched in 2023, this is **the world's first liquid hydrogen-powered ferry**, utilizing Ballard fuel cell systems. It operates in Norway, aligning with the country's goal for zero-emission fjord transport by 2026 [11].
- **Sea Change (USA)**: This 75-passenger catamaran began operations in the San Francisco Bay Area in 2024, using compressed hydrogen and fuel cells. It is **the first commercial hydrogen ferry in the U.S.** [12].
- **CMB's Hydroville and HydroBingo**: Following the Hydroville pilot, CMB launched HydroBingo in Japan, an 80-passenger ferry with dual hydrogen-diesel engines, showcasing a hybrid approach. Future projects include hydrogen fuel cell cruise ships and cargo vessels, with ammonia also being a key focus for decarbonizing large ocean-going ships.

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5. FUTURE EXPANSION: INVESTMENTS AND STRATEGIC INITIATIVES

The transition to hydrogen in transport and industry is supported by significant investments and strategic planning from corporations and governments.

5.1 Automotive Commitments

- **Honda**: Launching a CR-V Fuel Cell SUV in 2024 with a plug-in battery feature, produced in Ohio. The new fuel cell system, co-developed with GM, is also intended for heavy trucks and stationary power.
- **Toyota**: Beyond the Mirai, Toyota is developing fuel cell buses, pickup trucks (Hilux prototype), and actively partnering to build refueling infrastructure globally.
- **Hyundai**: Plans to introduce next-generation, lower-cost fuel cell systems by 2025 for diverse applications including cars, trucks, construction machinery, and ships, backed by a **$6–7 billion investment by 2030** [13].

5.2 Energy Sector Involvement

Traditional energy companies like Shell, TotalEnergies, and BP are investing in hydrogen refueling station networks and green hydrogen production facilities (e.g., Shell's Holland Hydrogen I). Amazon has secured agreements for green hydrogen to power its logistics fleet.

5.3 Emerging Players

Companies such as Nikola Motors and Hyzon Motors are developing and deploying hydrogen fuel cell trucks, contributing to the expansion of hydrogen-powered commercial fleets and supporting infrastructure.

5.4 Government Support and Targets

- **European Union**: The EU Hydrogen Strategy includes ambitious targets for FCEV deployment and **over 1,000 hydrogen fueling stations by 2030**, supported by funding programs like IPCEI.
- **Japan and South Korea**: Aim for millions of FCEVs by the 2030s, with substantial subsidies and infrastructure development plans.
- **China**: Leads in hydrogen bus and truck deployment, with around **10,000 FCEVs by 2022** and a target of **50,000 by 2025** [14]. China became the largest market for new hydrogen vehicles in 2023 [2].

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5.5 Advancements in Aviation and Shipping

Beyond current trials, aerospace companies like Boeing are studying hydrogen, and startups like ZeroAvia are testing hydrogen fuel cells in smaller aircraft. In shipping, Maersk is investing in green methanol-fueled ships as a transition towards hydrogen/ammonia, and major ports are developing hydrogen bunkering capabilities.

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6. CONCLUSION

The real-world examples presented herein underscore hydrogen's growing significance as a versatile and effective solution for decarbonizing diverse transportation and industrial sectors. From passenger vehicles and urban buses to heavy-duty trucks, trains, maritime vessels, and industrial equipment, hydrogen technologies — both fuel cells and combustion engines — are demonstrating practical viability and tangible emissions reductions.

Supported by substantial private and public investment, and driven by increasingly ambitious climate targets, these pioneering applications are paving the way for hydrogen to play a critical role in achieving a global zero-emission future, complementing electrification efforts and extending clean energy solutions to the most challenging applications.

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